



Course Name:	DSIP	Semester:	IV
Date of Performance:	02 / 02 / 2026	DIV/ Batch No:	A1
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Experiment No: 2

Title: Represent discrete time signals and perform different operations on them.

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Objective: To familiarize the beginner to MATLAB by introducing the basic features and commands of the program.

Expected Outcome of Experiment:

CO	Outcome
CO1	Identify various discrete time signals and systems and perform signal Manipulation

Books/ Journals/ Websites referred:

1. <http://www.mathworks.com/support/>
2. www.math.mtu.edu/~msgocken/intro/intro.html
3. www.mccormick.northwestern.edu/docs/efirst/matlab.pdf
4. A.Nagoor Kani "Digital Signal Processing", 2nd Edition, TMH Education.

Pre Lab/ Prior Concepts:

Using MATLAB we can easily generate all basic functions such as unit step, ramp, growing and decaying exponential, etc. The various signals plotted in this program are Step signal, Ramp signal, Exponential signal etc

1. Unit Step Signal

The step signal is defined as

$$U[n] = k; \text{ if } n \geq 0 \\ = 0; \text{ otherwise}$$

When $k=1$ it is called as unit step signal.

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2. Ramp Signal

The ramp signal is defined

as $r[n] = n$; if $n \geq 0$

$= 0$; otherwise

3. Exponential Signal

The exponential signal is defined as

$X[n] = a^n$

When 'a' is greater than 1 it is **increasing** exponential

When 'a' is less than 1 it is **decaying** exponential.

4. Impulse Signal

The impulse signal is defined

as $d[n] = k$; if $n=0$

$= 0$; otherwise

When $k=1$ it is called as unit impulse

The functions used in this program are:

a. Ones

This function is used to create an array of all ones

Syntax: $Y = \text{ones}(m, n)$

Description:

$Y = \text{ones}(n)$ returns an n-by-n matrix of 1's.

An error message appears if n is not a scalar.

$Y = \text{ones}(m, n)$ or $Y = \text{ones}([m \ n])$ returns an m-by-n matrix of ones.

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b. Zeros

This function is used to create an array of all zeros

Syntax: $Y = \text{zeros}(m,n)$

Description:

$Y = \text{zeros}(n)$ returns an n-by-n matrix of 0's.

An error message appears if n is not a scalar.

$Y = \text{zeros}(m,n)$ or $Y = \text{ones}([m\ n])$ returns an m-by-n matrix of Zeros.

c. EXP

This function is used to plot exponential signals

Syntax: $Y = \exp(X)$

Description:

The exp function is an elementary function that operates element-wise on arrays. Its domain includes complex numbers.

$Y = \exp(X)$ returns the exponential for each element of

X. For complex, it returns the complex exponential.

Steps with Syntax for representation of above discrete time signals:

```
clc;
```

```
clear;
```

```
close all;
```

```
n = -10:10;
```

```
u = (n >= 0);           % Unit Step  
r = n .* (n >= 0);      % Ramp
```

```
a1 = 1.5;               % Increasing exponential  
x1 = (a1.^n) .* (n >= 0);
```



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```
a2 = 0.6; % Decreasing exponential
x2 = (a2.^n) .* (n >= 0);
```

```
d = zeros(size(n)); % Unit Impulse
d(n == 0) = 1;
```

```
figure;
stem(n, u, 'filled');
title('Unit Step Signal');
xlabel('n'); ylabel('u[n]');
grid on;
```

```
figure;
stem(n, r, 'filled');
title('Ramp Signal');
xlabel('n'); ylabel('r[n]');
grid on;
```

```
figure;
stem(n, x1, 'filled');
title('Increasing Exponential Signal');
xlabel('n'); ylabel('x_1[n]');
grid on;
```

```
figure;
stem(n, x2, 'filled');
title('Decreasing Exponential Signal');
xlabel('n'); ylabel('x_2[n]');
grid on;
```

```
figure;
stem(n, d, 'filled');
title('Unit Impulse Signal');
xlabel('n'); ylabel('\delta[n]');
grid on;
```

```
figure;

subplot(3,2,1);
stem(n, u, 'filled');
title('Unit Step');
xlabel('n'); ylabel('u[n]');
grid on;
```

```
subplot(3,2,2);
stem(n, r, 'filled');
title('Ramp');
```

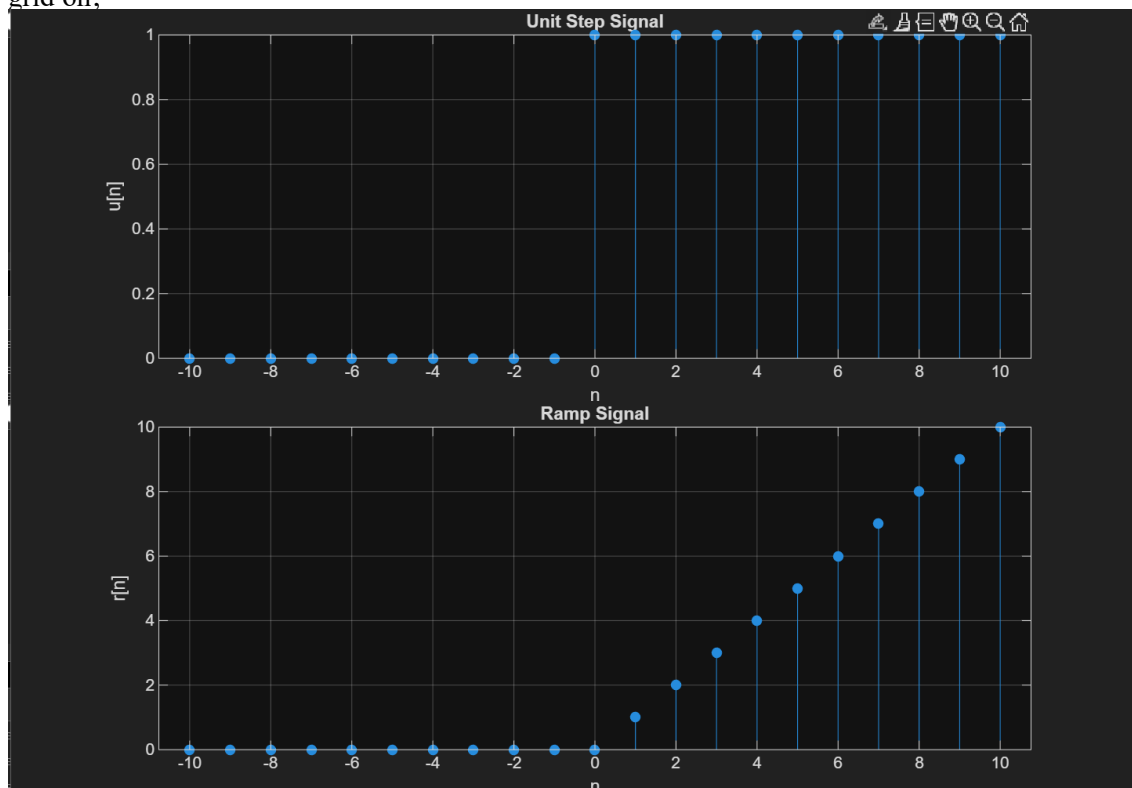
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```
xlabel('n'); ylabel('r[n]');
grid on;
```

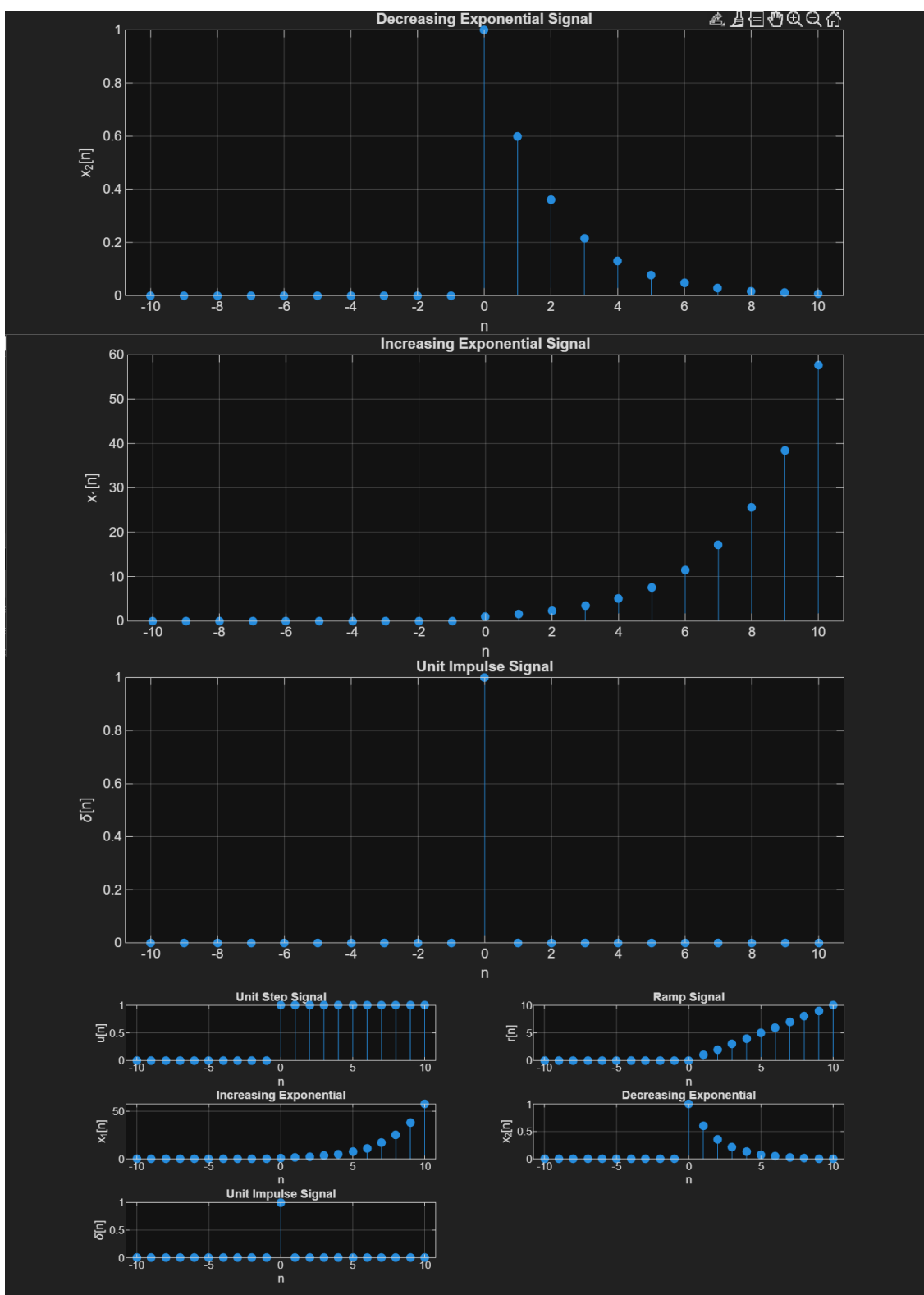
```
subplot(3,2,3);
stem(n, x1, 'filled');
title('Increasing Exp');
xlabel('n'); ylabel('x_1[n]');
grid on;
```

```
subplot(3,2,4);
stem(n, x2, 'filled');
title('Decreasing Exp');
xlabel('n'); ylabel('x_2[n]');
grid on;
```

```
subplot(3,2,5);
stem(n, d, 'filled');
title('Unit Impulse');
xlabel('n'); ylabel('\delta[n]');
grid on;
```



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Operations on Signals:

1. Addition of signals.
2. Subtraction of signals.
3. Multiplication of two signals.
4. Scaling – Upscaling & Downscaling.
5. Shift operation – Advance/Right shift & Delay/Left shift.

Steps with Syntax for representation of above operations on discrete time signals:

```
clc;
clear;
close all;

n = -10:10;

x1 = (n >= 0);    % Unit step signal
x2 = n .* (n >= 0); % Ramp signal

%% 1. Addition of Signals
add = x1 + x2;

%% 2. Subtraction of Signals
sub = x1 - x2;

%% 3. Multiplication of Signals
mul = x1 .* x2;

%% 4. Scaling of Signal
up_scale = 2 * x1; % Upscaling
down_scale = 0.5 * x1; % Downscaling

%% 5. Shift Operation
delay = (n-2 >= 0);
advance = (n+2 >= 0);

figure;

subplot(3,3,1);
stem(n, x1, 'filled');
title('Signal x1[n]');
grid on;

subplot(3,3,2);
stem(n, x2, 'filled');
title('Signal x2[n]');
grid on;

subplot(3,3,3);
```

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```

stem(n, add, 'filled');
title('Addition');
grid on;

subplot(3,3,4);
stem(n, sub, 'filled');
title('Subtraction');
grid on;

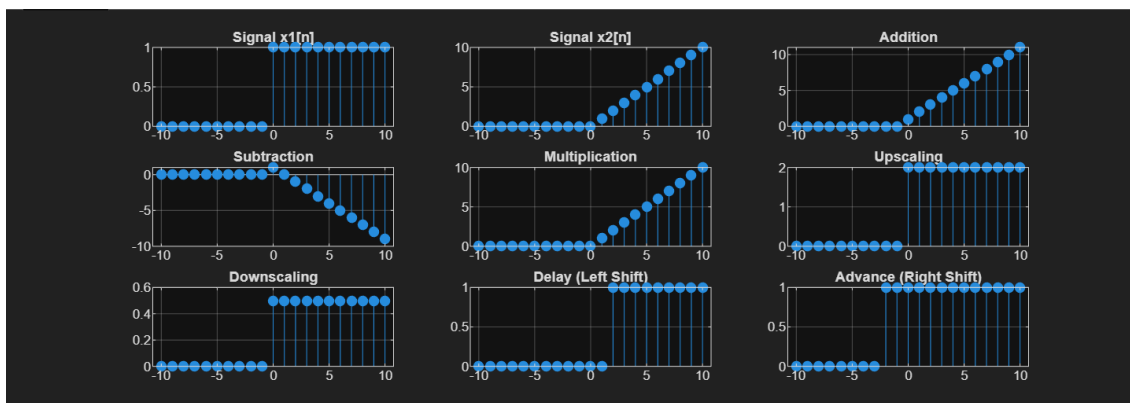
subplot(3,3,5);
stem(n, mul, 'filled');
title('Multiplication');
grid on;

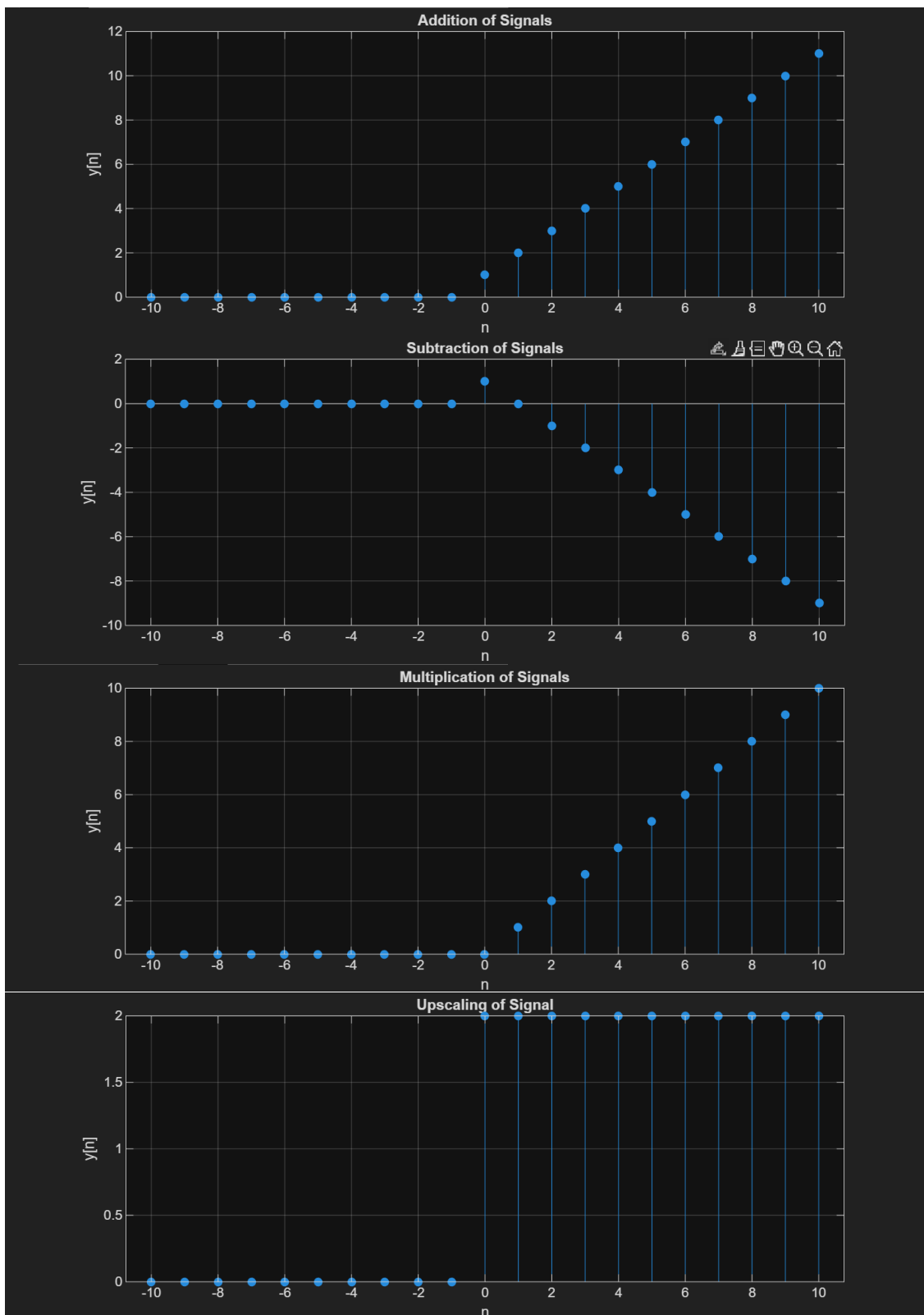
subplot(3,3,6);
stem(n, up_scale, 'filled');
title('Upscaling');
grid on;

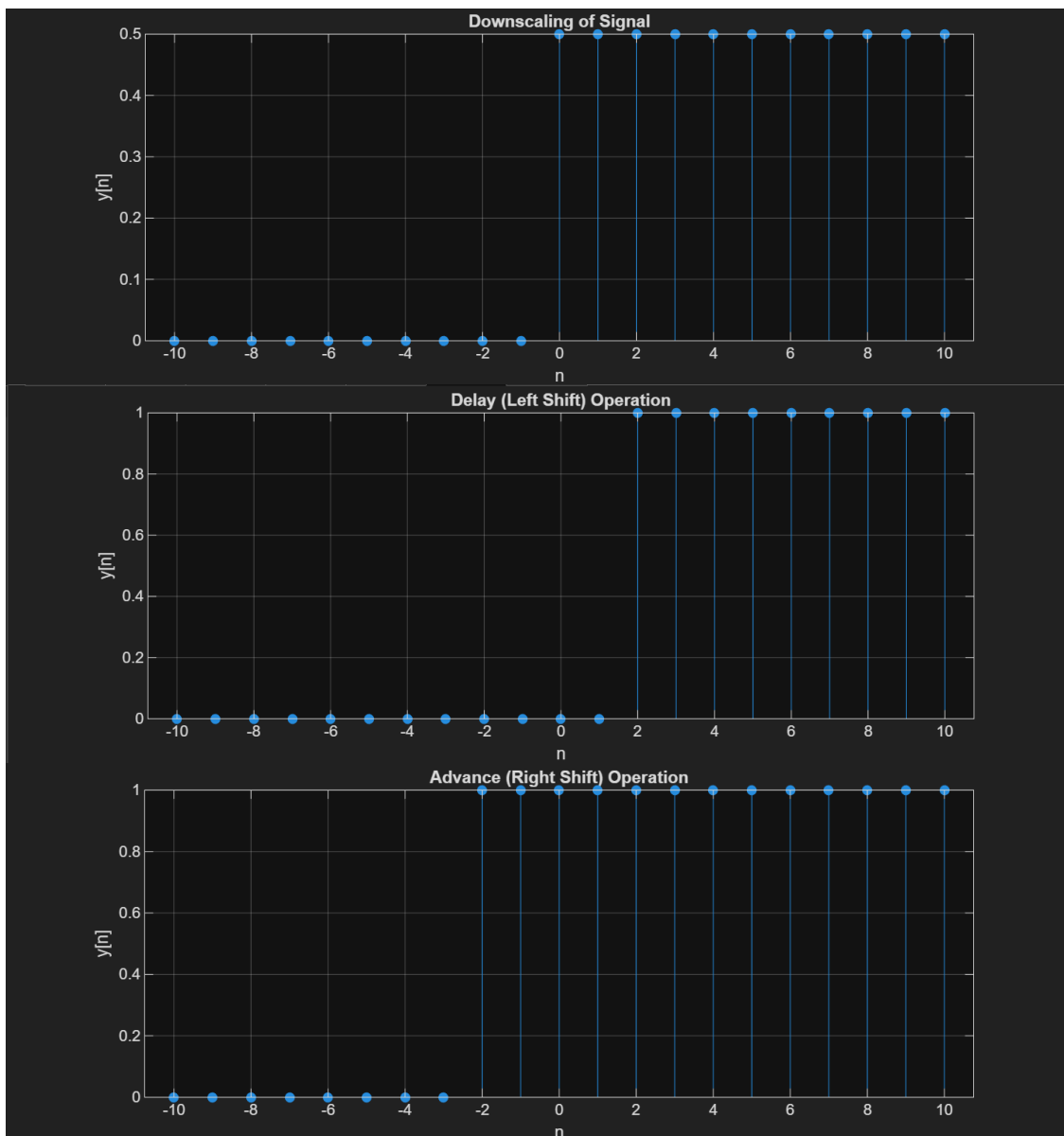
subplot(3,3,7);
stem(n, down_scale, 'filled');
title('Downscaling');
grid on;

subplot(3,3,8);
stem(n, delay, 'filled');
title('Delay (Left Shift)');
grid on;

subplot(3,3,9);
stem(n, advance, 'filled');
title('Advance (Right Shift)');
grid on;
  
```







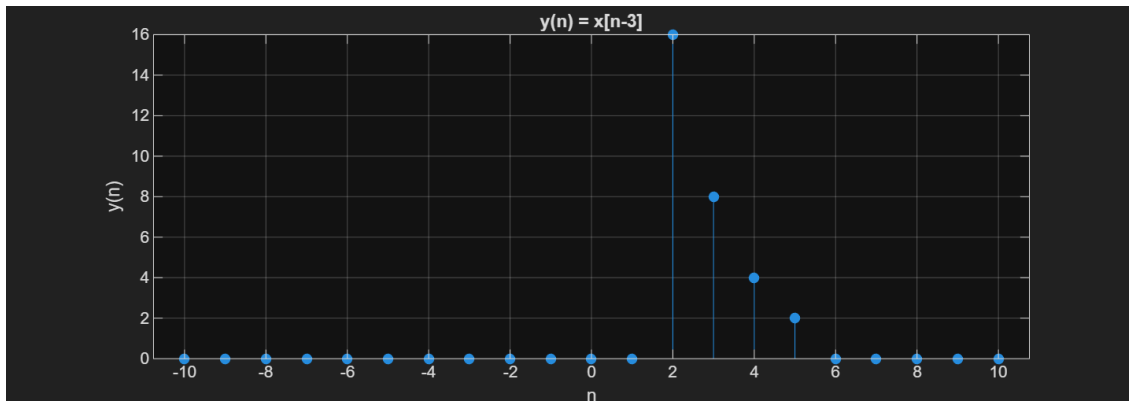
Conclusion:

In this experiment, various basic discrete-time signals such as unit step, ramp, impulse, and exponential signals were successfully generated and visualized using MATLAB. Fundamental signal operations including addition, subtraction, multiplication, scaling, and time shifting were performed on discrete-time signals. This experiment helped in understanding the representation, manipulation, and behavior of discrete signals in the time domain and provided hands-on familiarity with MATLAB commands used in Digital Signal Processing. The objectives of the experiment were achieved successfully.

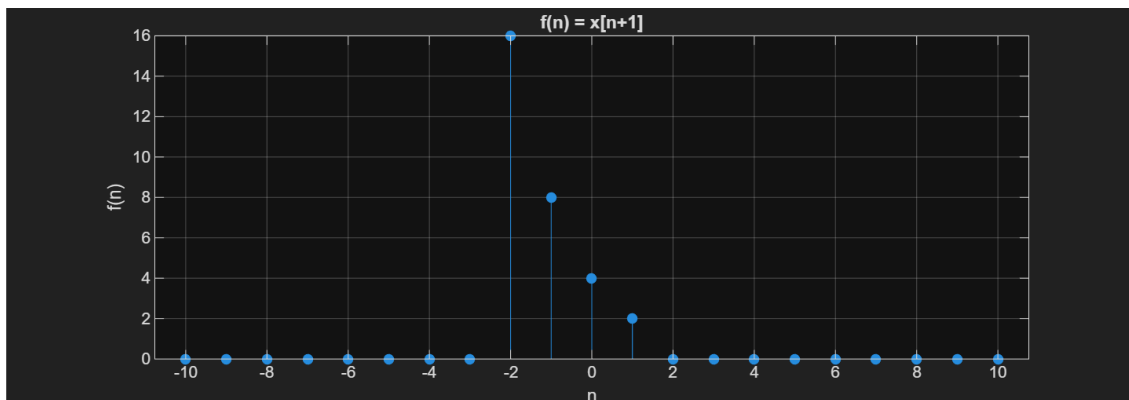
Post Lab Questions

1. Let $x(n) = 8(0.5)^n (u[n+1] - u[n-3])$. Sketch the following signals

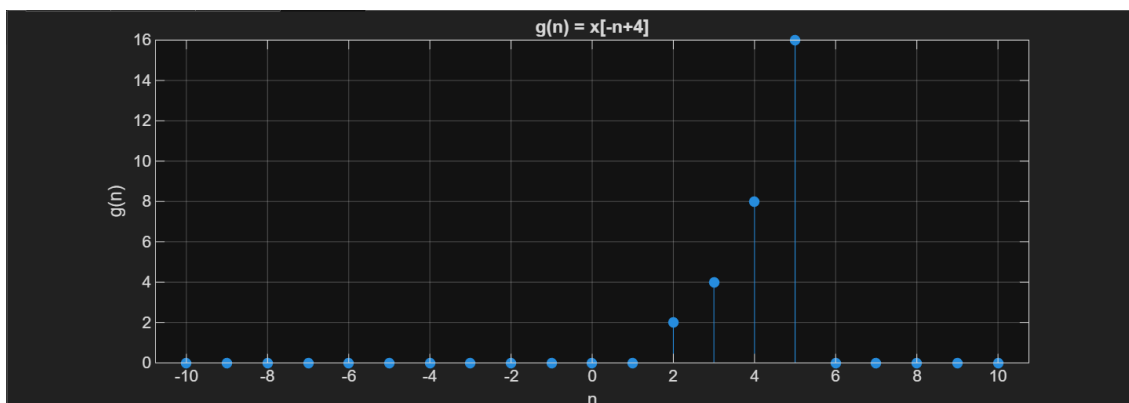
I. $Y(n) = [x-3]$



II. $F(n) = x[n+1]$



III. $G(n) = x[-n+4]$



2. The process of conversion of continuous time signal into discrete time signal is known as **Sampling**.
3. Which of the following is example of deterministic signal?
 - a. Step



- b. Ramp
 - c. Exponential
 - d. All of the above
4. For energy signals the energy will be finite and the average power will be zero.
5. In a signal $x(n)$, if 'n' is replaced by 'n/3' then it is called Time expansion.
6. The system $y(n) = \sin[x(n)]$ is
- a. Stable
 - b. BIBO stable
 - c. Unstable
 - d. None of the above